

CHAPTER 4: SYSTEM REQUIREMENTS

Functional, Non-Functional, Hardware, Software Specifications & Use Cases

4.1 Introduction

System requirements define the hardware, software, functional, and non-functional specifications required for the successful implementation of the proposed system. These requirements ensure that the platform performs efficiently, remains scalable, and provides a seamless experience to researchers and students. ResearchGraph AI is designed as a web-based application capable of integrating Artificial Intelligence, Knowledge Graphs, Vector Databases, and Large Language Models into a unified academic research platform.

4.2 Functional Requirements

- **FR1: User Search:** The system shall allow users to search research papers using natural language queries and support semantic search instead of simple keyword matching.
- **FR2: Research Paper Retrieval:** The system shall retrieve research papers from academic repositories such as OpenAlex and arXiv and display paper metadata including title, authors, publication year, journal, abstract, and DOI.
- **FR3: Semantic Search:** The system shall convert user queries into vector embeddings and retrieve the most semantically relevant paper chunks from the vector database.
- **FR4: Knowledge Graph Generation:** The system shall identify entities such as papers, authors, datasets, methods, and algorithms, automatically establish semantic relationships among these entities, and store them in Neo4j.
- **FR5: AI Chat Assistant:** The system shall answer research-related questions using an LLM, generate citation-supported responses, and summarize research papers.
- **FR6: Literature Review Generation:** The system shall automatically generate literature reviews from selected research papers, summarizing methodologies, datasets, advantages, and limitations.
- **FR7: Paper Comparison:** The system shall compare multiple research papers based on Dataset, Methodology, Algorithms, Accuracy, Advantages, and Limitations.
- **FR8: Research Gap Identification:** The system shall identify unexplored research areas by analyzing trends and limitations across multiple papers.
- **FR9: Knowledge Graph Visualization:** The system shall display relationships among papers, authors, datasets, and models using an interactive graph.
- **FR10: User Profile:** The system shall allow users to save papers, maintain search history, and store generated literature reviews.

4.3 Non-Functional Requirements

Attribute	Specification & Requirement
Performance	Search results retrieved within seconds; graph rendering loads efficiently; AI response generation latency minimized.
Scalability	Architecture supports continuous indexing of newly published papers and seamless expansion of Neo4j Knowledge Graph.
Reliability	Provides highly accurate retrieval, consistent paper metadata, and 100% verifiable citations.
Availability	Accessible through modern web browsers with high backend service uptime.
Security	User authentication protects personal workspaces; secure HTTPS connections for API endpoints; confidential user data.
Usability	Intuitive UI requiring no specialized technical skills; simple module navigation.
Maintainability	Modular architecture facilitating modular model upgrades and new academic API data source additions.
Portability	Cross-platform compatibility (Windows, Linux, macOS) via web browser; deployable on cloud platforms.

4.4 Hardware Requirements

Component	Development Specification	Production Server Specification
Processor / CPU	Intel Core i5 (or equivalent)	Multi-core Cloud Server (8+ vCPU)
RAM	Minimum 8 GB (16 GB Recommended)	16 GB RAM or Higher
Storage	20 GB Free Disk Space	High-Speed SSD Storage
Network	Stable Broadband Connection	High-Speed Internet Line

4.5 Software Requirements

Software Tool / Tech	Purpose in System
Operating System	Windows 10/11, Linux (Ubuntu/Debian)
Programming Languages	Python 3.10+, TypeScript
Frontend Framework	React + Tailwind CSS
Backend Framework	FastAPI
Vector Database	ChromaDB
Graph Database	Neo4j (Neo4j Aura / Local Enterprise)
Relational Database	PostgreSQL
AI Orchestration	LangChain
PDF Processing	PyMuPDF (Fitz)
Embedding Model	BAAI/BGE-large-en-v1.5 / E5
Large Language Model	Gemini / Llama 3
IDE & Version Control	Visual Studio Code, Git & GitHub

4.6 Actors & 4.7 Use Cases

- **Researcher:** Primary user who executes semantic searches, compares papers, generates literature reviews, views knowledge graphs, and chats with the AI assistant.
- **Administrator:** System administrator responsible for database maintenance, repository indexing pipelines, monitoring performance, and managing system configurations.

4.8 Use Case Diagram (Text Representation)

